Practical exercises — percentage change and simple interest

A Cambridge insert: increase, decrease, and the reverse question that catches everybody.

LESSON 131–132 | 2 × 40 MIN | MATEMATIKA 7, AMALIY MASHQLAR | S8 10

LESSON CLOCK

15' | 20' | 25' | 20' | 5'

One multiplication, not two steps	15 min
Finding the change	20 min
Simple interest	25 min
The reverse question	20 min
Homework	5 min

LEARNING OBJECTIVES

- Increase and decrease a quantity by a given percentage in one multiplication.
- Find a percentage change from the old and new values.
- Calculate simple interest over several years.
- Find the original amount when the changed one is given.

TEXTBOOK

National	pp. 387–392
Cambridge	Stage 8, pp. 96–104
Workbook	Exercise 10.2

01 Explanation

One multiplication, not two steps

An increase of *15%* multiplies by *1.15*; a decrease of *15%* multiplies by *0.85*. Finding the part and then adding it is the same thing done twice.

$$\text{new value} = \text{old value} \times \left(1 \pm \frac{p}{100}\right)$$

CHANGE	MULTIPLIER	*4000* BECOMES
up *15%*	*1.15*	*4600*
down *15%*	*0.85*	*3400*
up *8%*	*1.08*	*4320*
down *40%*	*0.6*	*2400*

UP 10% THEN DOWN 10% IS NOT BACK WHERE YOU STARTED

$100 \cdot 1.1 \cdot 0.9 = 99$. The two percentages are taken of different amounts, so they do not cancel — a fact every discount advertisement relies on.

Finding the change

$$\text{percentage change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100\%$$

OLD	NEW	CHANGE	PERCENTAGE
50	65	$+15$	$+30\%$
80	60	-20	-25%
200	250	$+50$	$+25\%$
250	200	-50	-20%

ALWAYS DIVIDE BY THE *OLD* VALUE

The last two rows are the same pair of numbers and give different percentages, because the base changed. A rise of 25% is undone by a fall of 20%, not of 25%.

Simple interest

$$I = \frac{\mathrm{P} \cdot \mathrm{r} \cdot \mathrm{t}}{100}$$

with P the principal, r the yearly rate as a percentage and t the number of years.
Simple interest is calculated on the original sum every year.

P	R	T	INTEREST	TOTAL
$1\,000\,000$	12%	1	$120\,000$	$1\,120\,000$
$1\,000\,000$	12%	3	$360\,000$	$1\,360\,000$
$500\,000$	8%	2	$80\,000$	$580\,000$

SIMPLE, NOT COMPOUND

Here the second year earns interest on $1\ 000\ 000$ again, not on $1\ 120\ 000$. Compound interest, where it does, waits until Grade 9 — but the difference is worth stating now.

The reverse question

"After a rise of 20% the price is $60\ 000$. What was it before?" The $60\ 000$ is **not** the base: divide by the multiplier rather than taking 20% of it.

GIVEN	WRONG	RIGHT	ANSWER
after $+20\%$ it is $60\ 000$	$60\ 000 - 12\ 000$	$60\ 000 \div 1.2$	$50\ 000$
after -20% it is $60\ 000$	$60\ 000 + 12\ 000$	$60\ 000 \div 0.8$	$75\ 000$
after $+25\%$ it is 500	$500 - 125$	$500 \div 1.25$	400

CHECK BY GOING FORWARDS

$50\ 000 \cdot 1.2 = 60\ 000$ ✓. The reverse question is the one most often got wrong in the whole of percentages, and the forward check takes five seconds.

02 Worked examples

EXAMPLE 1 **A price of 4000 sum rises by 15%. Find the new price.**

1. The multiplier is 1.15.
2. $4000 \cdot 1.15$
3. $= 4600$

ANSWER

4600 sum

EXAMPLE 2 **$1\ 000\ 000$ sum is invested at 12% simple interest for 3 years. Find the interest and the total.**

1. $I = \dfrac{1\ 000\ 000 \cdot 12 \cdot 3}{100}$

2. $= 360\ 000$

   Each year earns $120\ 000$.

3. Total $= 1\ 360\ 000$.

ANSWER

$360\ 000$ interest, $1\ 360\ 000$ in all

EXAMPLE 3 **After a rise of 20% a price is $60\ 000$ sum. What was it before?**

1. The old price was multiplied by 1.2.

2. $60\ 000 \div 1.2$

   Divide, do not subtract.

3. $= 50\ 000$

4. Check: $50\ 000 \cdot 1.2 = 60\ 000$ ✓

ANSWER

$50\ 000$ sum

03 Interactive model

Ask the class whether a 20% rise followed by a 20% fall returns the original price; the near-unanimous “yes” makes the arithmetic that follows memorable.

Multiplier, change, or reverse?

The multiplier for a rise of 15% is:

- 0.15
- 1.15
- 15
- 0.85

The multiplier for a fall of 15% is:

- 0.85
- 1.15
- 0.15
- -15

4000 up 15% is:

- 4150
- 4600
- 4015
- 3400

From 50 to 65 is a change of:

- 15%
- 23%
- 30%
- 130%

From 80 to 60 is a change of:

- -20%
- -25%
- -33%
- -75%

$1\,000\,000$ at 12% simple for 3 years earns:

- $120\,000$
- $360\,000$
- $404\,928$
- $36\,000$

After $+20\%$ a price is $60\,000$; before it was:

- $48\,000$
- $50\,000$
- $72\,000$
- $40\,000$

Up 10% then down 10% gives:

- the same value
- 99% of it
- 101% of it
- 90% of it

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Percentage	Foiz	Процент
Increase	Ortish	Увеличение
Decrease	Kamayish	Уменьшение
Multiplier	Ko‘paytuvchi	Множитель
Percentage change	Foizli o‘zgarish	Процентное изменение
Simple interest	Oddiy foiz	Простые проценты
Principal	Asosiy summa	Основная сумма
Original value	Boshlang‘ich qiymat	Первоначальное значение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

To increase by *8%* multiply by:

- *0.08*
- *1.08*
- *8*
- *0.92*

To decrease by *40%* multiply by:

- *0.4*
- *0.6*
- *1.4*
- *0.96*

Percentage change divides by:

- the new value
- the old value
- the difference
- *100*

Simple interest for t years is:

- $\frac{Prt}{100}$
- $\frac{Pr}{100}$
- $P(1 + r)^t$
- Prt

After -20% a price is $60\ 000$; before it was:

- $72\ 000$
- $75\ 000$
- $48\ 000$
- $80\ 000$

A rise of 25% is undone by a fall of:

- 25%
- 20%
- 30%
- 75%

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 4000 up 15%

ANSWER 4600

2. 4000 down 15%

ANSWER 3400

3. 4000 up 8%

ANSWER 4320

4. 4000 down 40%

ANSWER 2400

5. The multiplier for $+12\%$

ANSWER 1.12

6. The multiplier for -7%

ANSWER 0.93

7. 20% of 350

ANSWER 70

Medium · the standard question

7 PROBLEMS

1. From 50 to 65

   ANSWER $+30\%$

2. From 80 to 60

   ANSWER -25%

3. $1\,000\,000$ at 12% simple, 3 years: the interest

   ANSWER $360\,000$

4. $500\,000$ at 8% simple, 2 years: the total

   ANSWER $580\,000$

5. After $+20\%$ it is $60\,000$: before

   ANSWER $50\,000$

6. After -20% it is $60\,000$: before

   ANSWER $75\,000$

7. Up 10% then down 10% from 100

   ANSWER 99

Hard · stretch and reason

7 PROBLEMS

1. A rise of 25% is undone by a fall of

   ANSWER 20%

2. After two rises of 10% from 200

   ANSWER 242

3. A price falls 30% to $49\ 000$: the original

   ANSWER $70\ 000$

4. At 10% simple, how long until P doubles?

   ANSWER 10 years

5. P at 15% simple for 4 years: the total as a multiple of P

   ANSWER $1.6P$

6. A shop raises a price 20% then advertises 20% off: the net change

   ANSWER -4%

7. Interest of $90\ 000$ on $750\ 000$ over 2 years: the rate

   ANSWER 6%

07 Homework

Homework — 5 tasks

For every reverse question, check your answer by going forwards.

1. Increase $25\ 000$ by 18% and decrease it by 18%.
2. A price rises from $40\ 000$ to $46\ 000$. Find the percentage change.
3. Find the simple interest on $2\ 000\ 000$ at 9% for 4 years.
4. After a fall of 25% a price is $90\ 000$. What was it before?
5. Show that a rise of 50% followed by a fall of 50% loses a quarter of the original.